

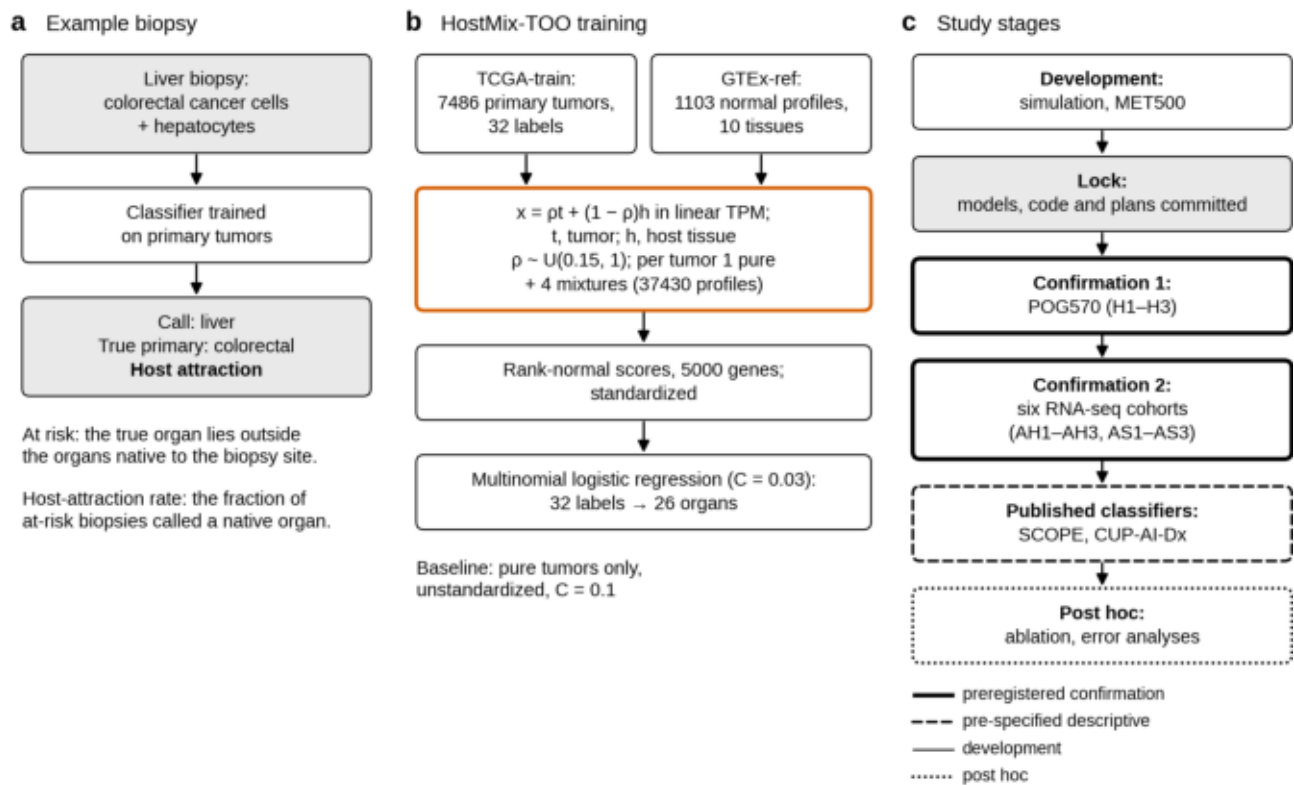


Figure 1. Host attraction, mixture-augmented training and study design

a A metastatic biopsy contains tumor cells and normal cells of the host organ. A classifier trained on primary tumors can name the host organ, here liver for a colorectal cancer metastasis in the liver; we call this error host attraction. A biopsy is at risk when its true organ lies outside the organs native to the biopsy site, and the host-attraction rate is the fraction of at-risk biopsies predicted as a native organ. b HostMix-TOO training. Each TCGA-train tumor profile is used once unchanged and in four in silico mixtures (orange border) with a normal profile of one of ten GTEx host tissues, mixed in linear TPM with tumor RNA fraction p . Within-sample rank-normal scores of 5000 genes are standardized and used as the input of a multinomial logistic regression ($C = 0.03$) with 32 tumor labels, whose probabilities are summed to 26 organ groups. The baseline classifier uses the same tumors and features without mixtures or standardization ($C = 0.1$). c Study stages. Simulated mixtures and MET500 were used for development (thin border). The baseline and HostMix-TOO were then locked (shaded box) and compared in two internally preregistered confirmations (thick borders). The comparison with published classifiers was specified in advance and is descriptive (dashed border); the ablation and the error analyses were post hoc (dotted border).

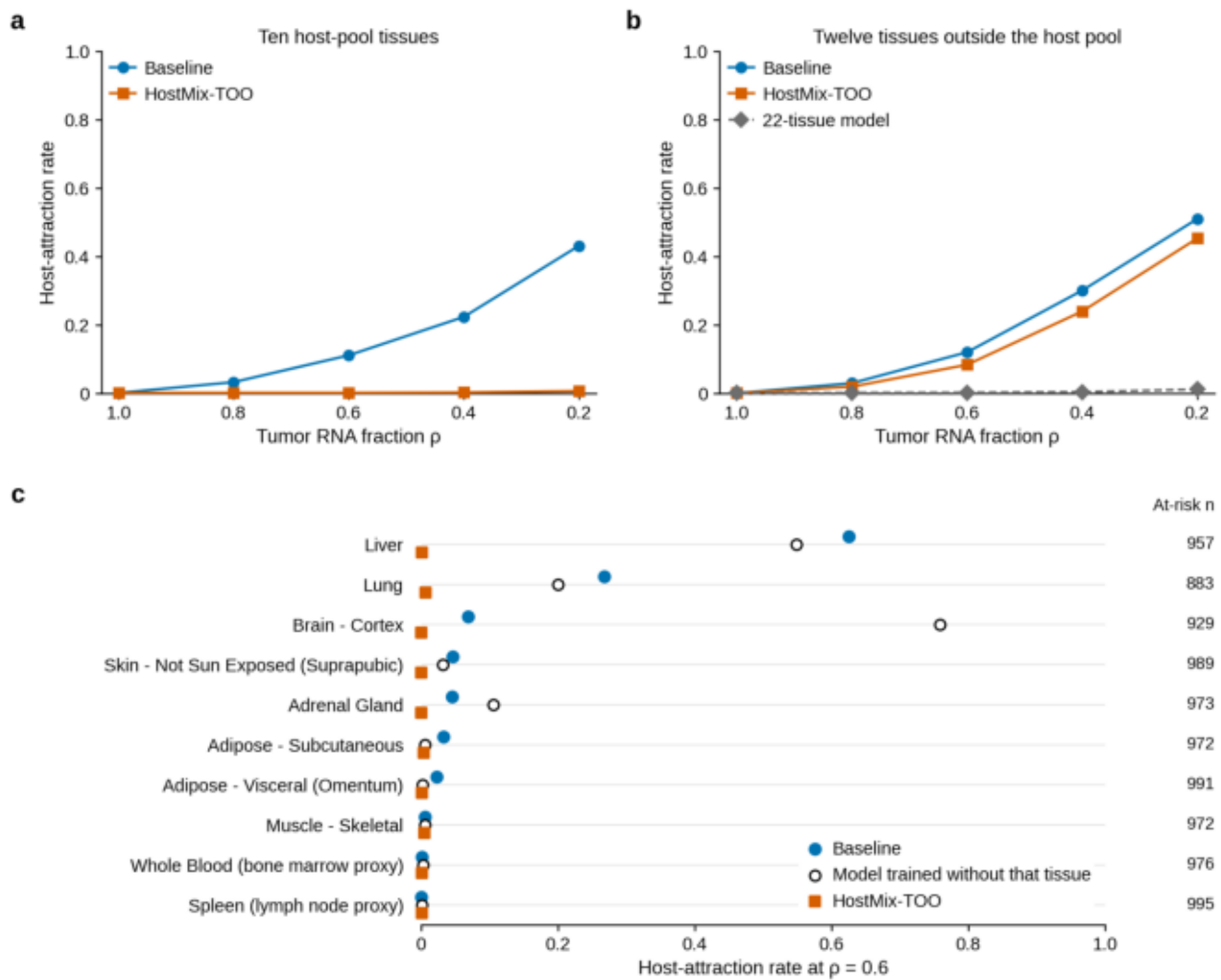


Figure 2. Host attraction in simulated tumor–host mixtures

Held-out TCGA tumors ($n = 1000$) were mixed with GTEx-sim samples at tumor RNA fraction p ; within each tissue, these donors had not been used for training. **a** Mean host-attraction rate across the ten tissues of the training host pool for the baseline and HostMix-TOO. **b** Mean host-attraction rate across twelve tissues outside the host pool for the baseline, HostMix-TOO and an exploratory model trained with all 22 tissues. Rates are computed over tumors whose organ is not native to the site that the tissue represents and are averaged over tissues without weighting; values per tissue are in Additional file 2. $p = 1$ corresponds to pure tumors. **c** Host-attraction rate at $p = 0.6$ on each tissue of the host pool for the baseline, HostMix-TOO and the leave-one-host-out model trained without that tissue; rows are ordered by the baseline rate, markers within a row are offset vertically for legibility, and the number of at-risk tumors for each tissue is given on the right. Spleen and whole blood represent lymph node and bone marrow sites. The simulations were part of development and are descriptive.

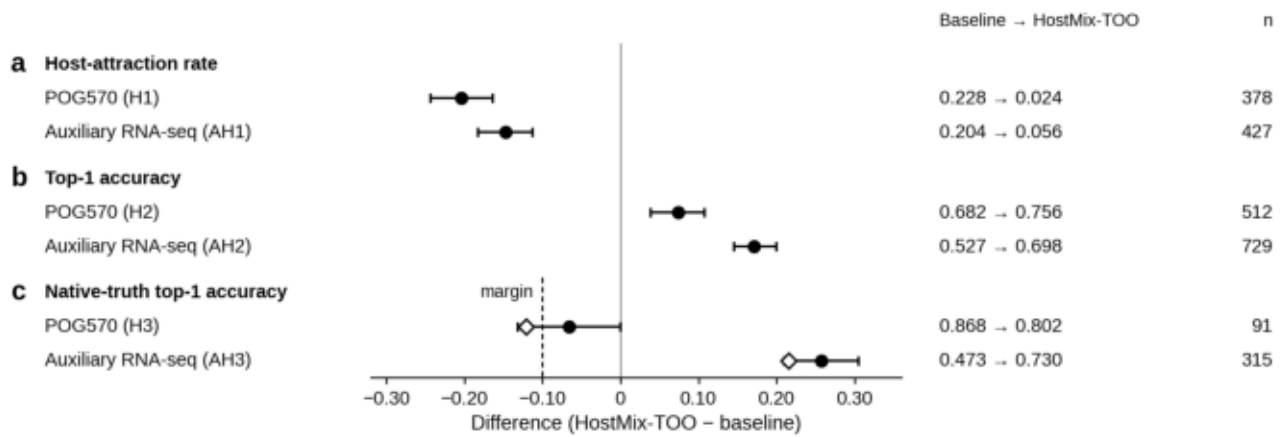


Figure 3. Preregistered comparisons of HostMix-TOO with the baseline classifier

Differences, HostMix-TOO minus baseline, with two-sided 95% patient-bootstrap percentile intervals in POG570 (H1–H3) and in the six auxiliary RNA-seq cohorts (AH1–AH3). a Host-attraction rate in the at-risk set. b Top-1 organ accuracy in the evaluation set. c Top-1 accuracy in the native-truth set, that is, biopsies taken from the organ of origin; the dashed line marks the non-inferiority margin of -0.10 , and the open diamond shows the one-sided 95% lower bound. The columns on the right give the baseline and HostMix-TOO estimates and the size of each analysis set. H1, H2, AH1 and AH2 were tested with exact one-sided McNemar tests. The secondary hypotheses are reported in Table 2.

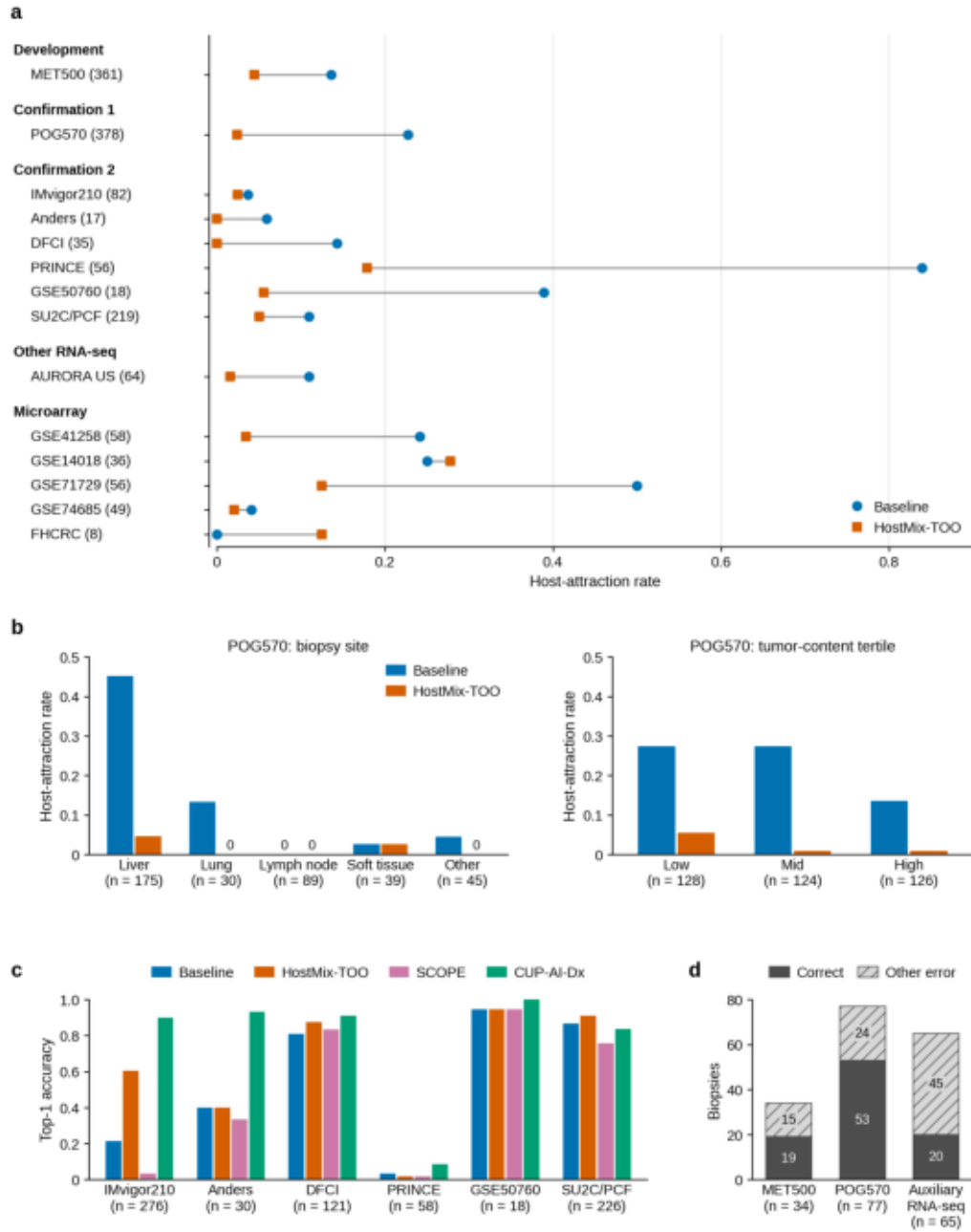


Figure 4. Host attraction and accuracy by cohort and stratum

a Host-attraction rate of the baseline and HostMix-TOO in each evaluation cohort, grouped by the role of the cohort; AURORA US (other RNA-seq) and the microarray cohorts are descriptive. Each line joins the two classifiers within one cohort, and the at-risk n in parentheses matches Table 1. b POG570 host-attraction rate by biopsy-site group and by tertile of tumor content (pre-specified descriptive); n is the number of at-risk biopsies in each group, and 0 marks a rate of zero. c Top-1 accuracy of the baseline, HostMix-TOO, SCOPE and CUP-AI-Dx in each auxiliary RNA-seq cohort, with the evaluation n (post hoc decomposition of Table 3). d At-risk biopsies whose baseline host-attraction error was removed by HostMix-TOO, divided into correct calls and other errors, in MET500 ($n = 34$), POG570 ($n = 77$) and the auxiliary cohorts ($n = 65$) (post hoc); the numbers in the bars are counts.

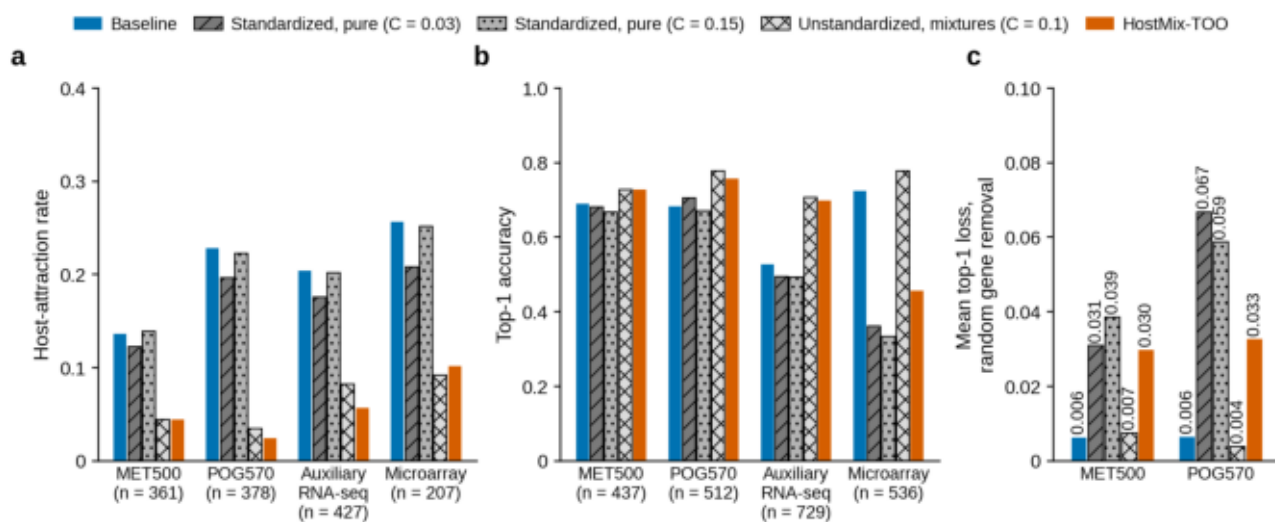


Figure 5. Post hoc ablation separating mixtures from feature standardization and regularization

a Host-attraction rate in the at-risk sets and b top-1 accuracy in the evaluation sets of MET500, POG570, the six auxiliary RNA-seq cohorts and the five microarray cohorts (n under each cohort) for five logistic models: the baseline (pure tumors, unstandardized, $C = 0.1$); two standardized models trained on pure tumors ($C = 0.03$ and $C = 0.15$); an unstandardized model trained with the HostMix-TOO mixtures ($C = 0.1$); and HostMix-TOO (mixtures, standardized, $C = 0.03$). c Mean loss of top-1 accuracy in MET500 and POG570 when as many randomly chosen classifier genes were removed as each of four microarray platforms lacks (four platforms, ten draws each). Separate plans for the RNA-seq cohorts and for the microarray and missing-gene extensions, each with a reading rule, were committed before the models were applied; all results are post hoc. Shares of the differences attributable to the mixtures are given in Additional file 1: Tables S8 and S9.

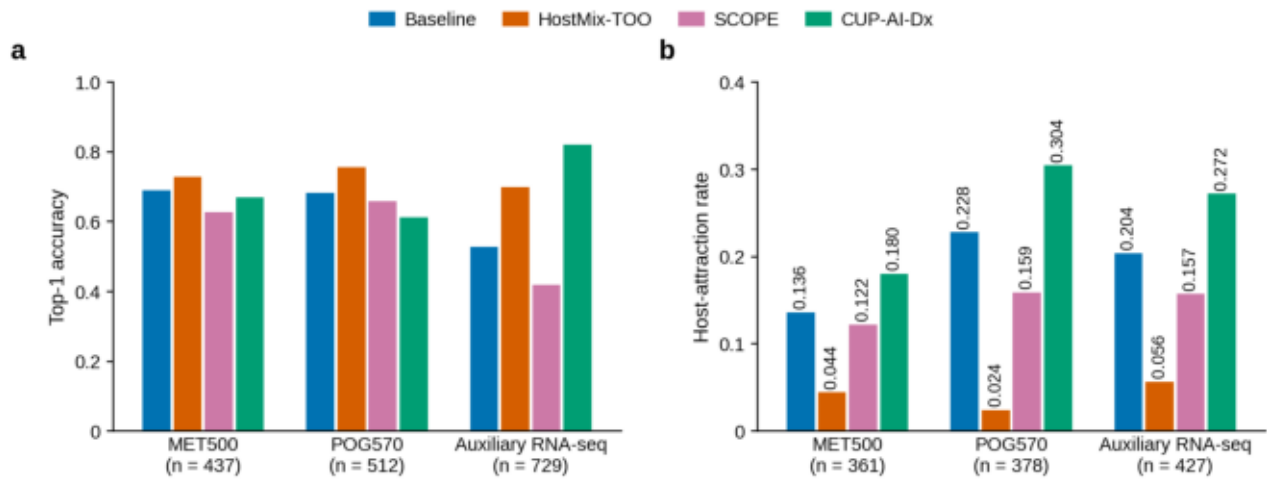


Figure 6. Host attraction and accuracy of two published classifiers

a Top-1 organ accuracy in the evaluation sets and b host-attraction rate in the at-risk sets of the baseline, HostMix-TOO, SCOPE and CUP-AI-Dx in MET500 (n = 437; at risk, 361), POG570 (512; 378) and the six auxiliary RNA-seq cohorts (729; 427); the n under each cohort is the denominator of that panel, and the numbers above the bars in b are the rates. SCOPE and CUP-AI-Dx were run with the authors' code and models on locked input matrices. The comparison was specified before the runs and is descriptive; paired differences with bootstrap intervals are in Additional file 2. SCOPE calls of normal-tissue classes were counted as incorrect. MET500 was the development cohort for HostMix-TOO.

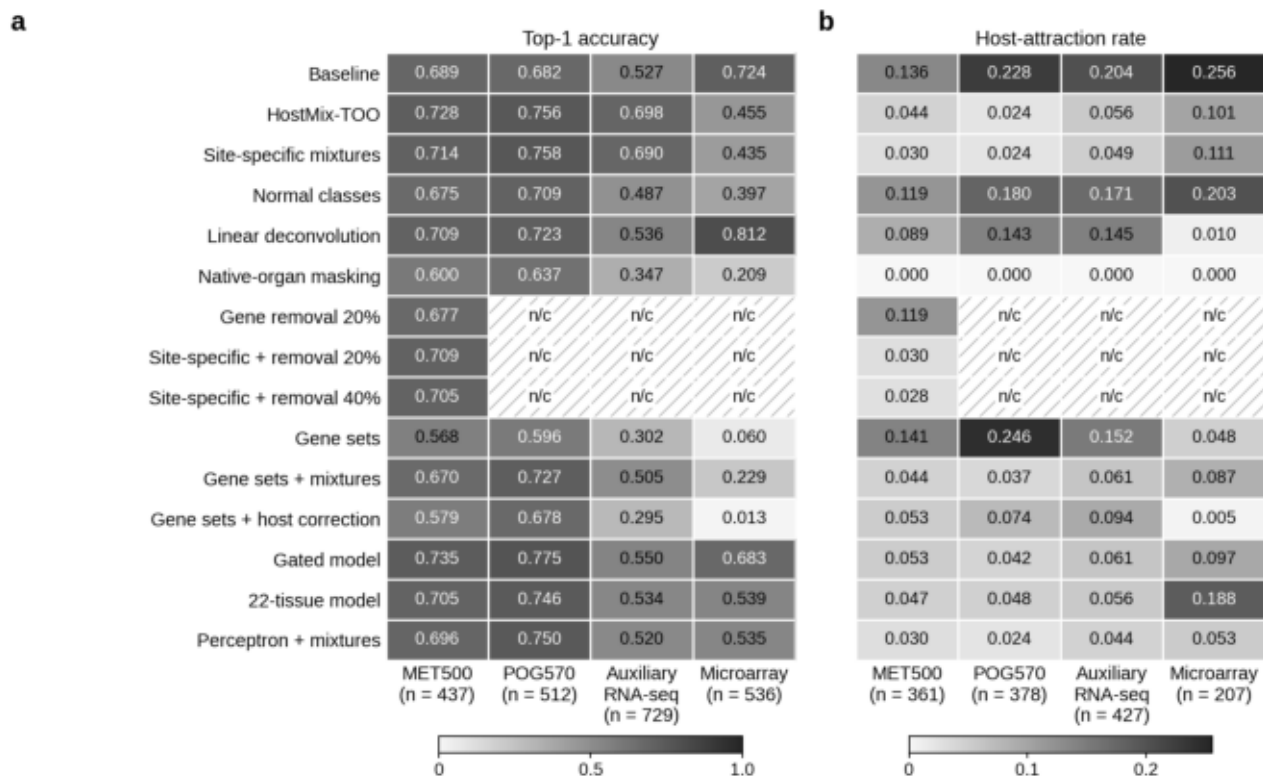


Figure S1. Accuracy and host attraction of the locked classifiers

Each cell is the top-1 organ accuracy (a) or the host-attraction rate (b) of one locked classifier. Columns are MET500, POG570, the six confirmatory RNA-seq cohorts, and the microarray cohorts. The MET500 column is exploratory. In the POG570 column, the baseline and HostMix-TOO cells are the two arms of the preregistered comparisons H1 and H2, and the other cells are pre-specified descriptive or exploratory; the status of every cell is given in Additional file 2. The auxiliary column is a post hoc aggregate of the confirmatory predictions on the six RNA-seq cohorts. The microarray column is a post hoc aggregate of the one-patient descriptive layer (top-1 on 536 evaluation biopsies; host-attraction rate on 207 at-risk biopsies). Shading runs from light to dark gray over 0 to 1 in a and over 0 to the largest rate in b; labels give the value to three decimals. n/c, not computed (hatched cells). Native-organ masking has a host-attraction rate of 0 by construction.

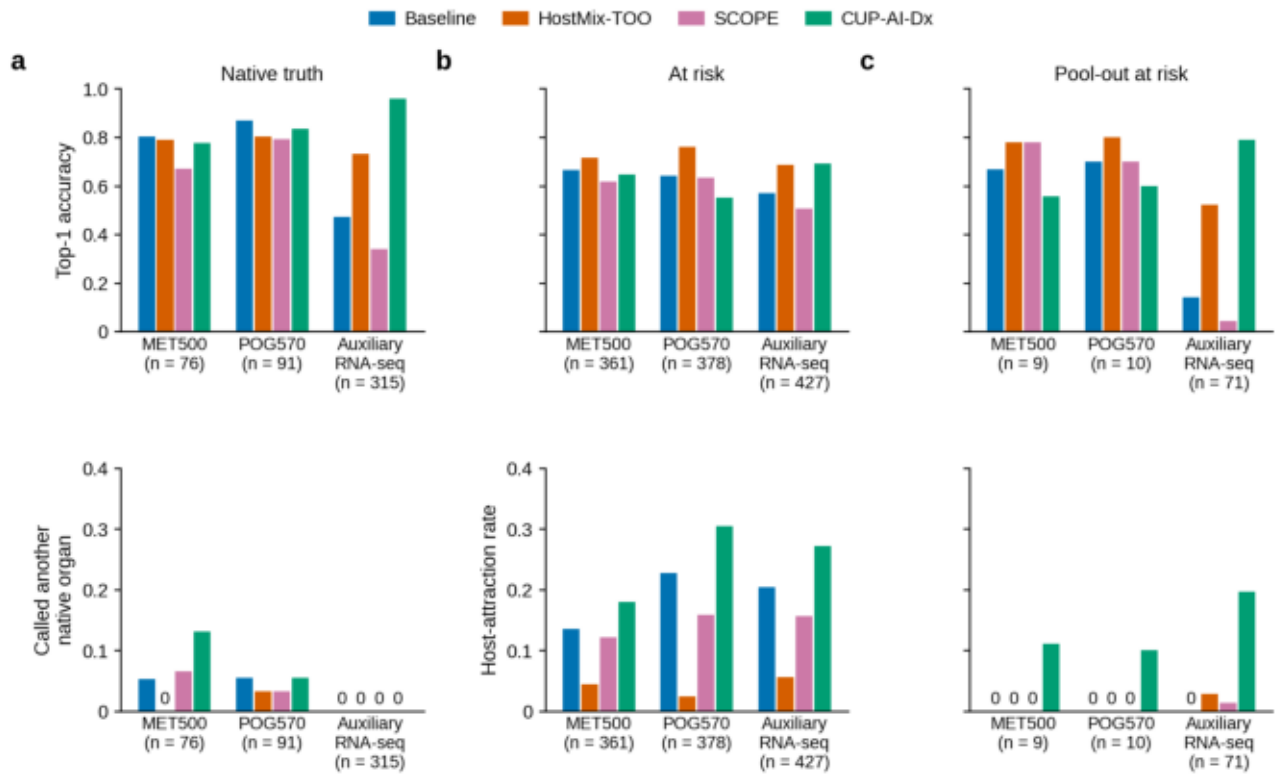


Figure S2. Accuracy and host attraction in three analysis sets

Top-1 accuracy (upper row) and a second metric (lower row) of the baseline, HostMix-TOO, SCOPE and CUP-AI-Dx in the native-truth set (a), the at-risk set (b) and the pool-out at-risk set (c), shown separately for MET500, POG570 and the auxiliary RNA-seq cohorts; n under each group is the size of that set. In b and c the lower row is the host-attraction rate. Host attraction is defined only for at-risk biopsies, so in a the lower row ("called another native organ") is the fraction of native-truth biopsies called a native organ of the biopsy site other than the true organ. A bar of height zero is labeled 0. This decomposition was made after both confirmations and is post hoc.

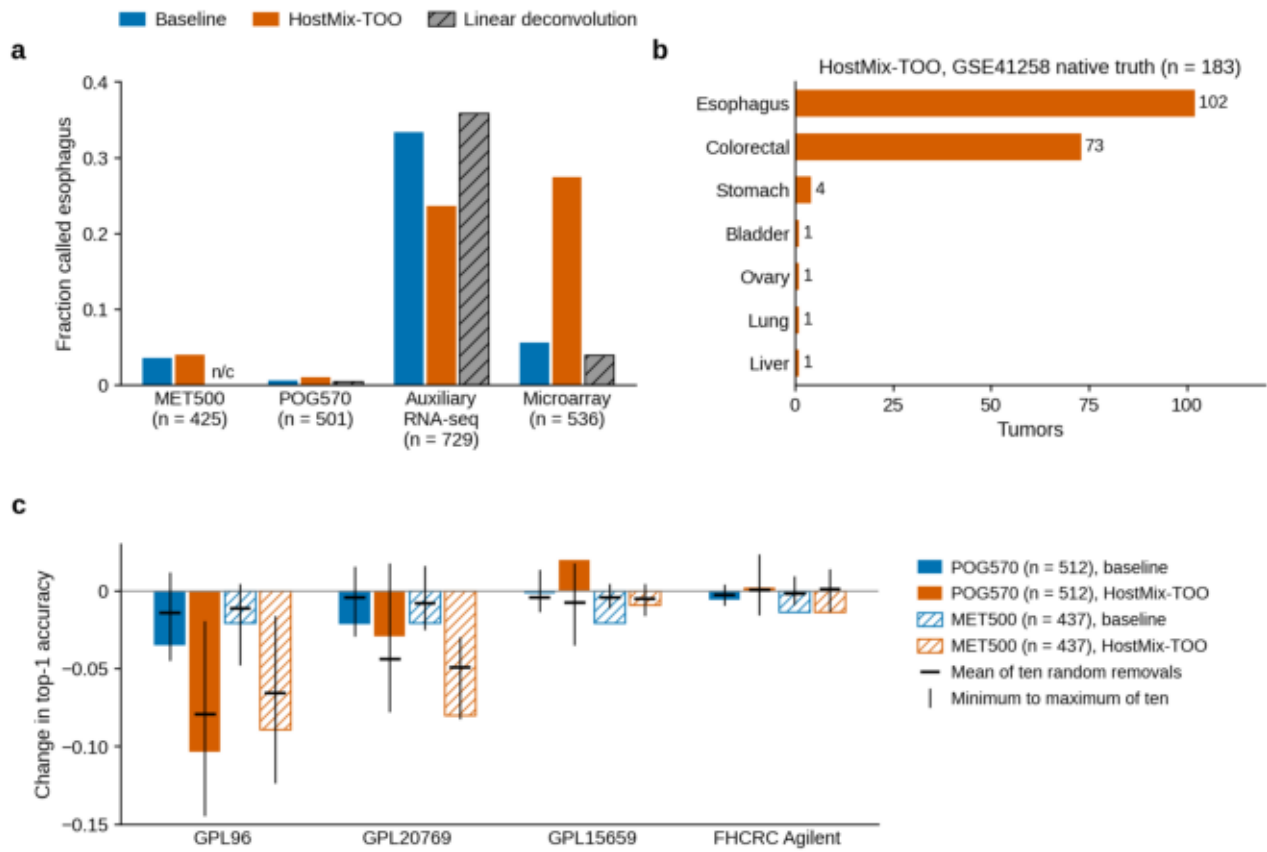


Figure S3. Esophagus calls and the effect of missing genes

a Fraction of biopsies whose true organ is not esophagus that were called esophagus, in MET500, POG570, the six auxiliary RNA-seq cohorts and the microarray cohorts; n is that denominator. For the microarray cohorts, the set has one sample per patient across the layer, as in Table 1 (n = 536). n/c, not computed: the esophagus fraction of linear deconvolution was not computed for MET500. b HostMix-TOO calls for the 183 GSE41258 native-truth colorectal tumors; 102 were called esophagus. c Change in top-1 accuracy after restricting POG570 and MET500 profiles to the genes of GPL96, GPL20769, GPL15659 or the FHCRC Agilent array; solid bars are POG570 and hatched bars MET500, colored by classifier. The horizontal mark is the mean change across ten random removals of the same number of classifier genes, and the vertical line spans the minimum to maximum of those ten changes. The missing-gene simulation is post hoc. Models were not retrained.

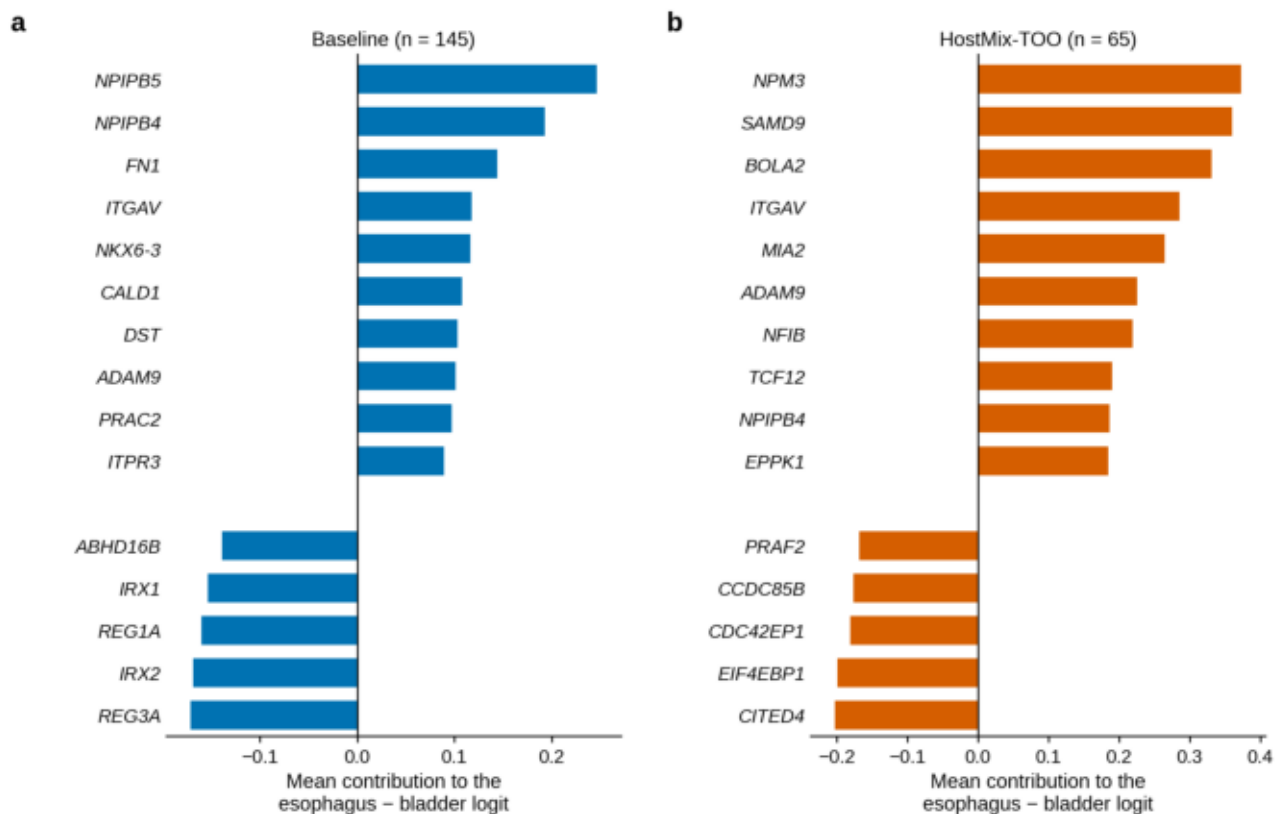


Figure S4. Genes contributing to esophagus calls in IMvigor210

Mean contribution of each gene to the difference between the esophagus and bladder logits for IMvigor210 bladder tumors in the native-truth set that were called esophagus: a baseline (n = 145), b HostMix-TOO (n = 65). The contribution is the coefficient difference multiplied by the model input, the rank-normal score for the baseline and the standardized score for HostMix-TOO, so values are not comparable between panels. Positive values favor esophagus. The ten largest positive and the five most negative mean contributions are shown. This analysis is post hoc and does not identify a mechanism.

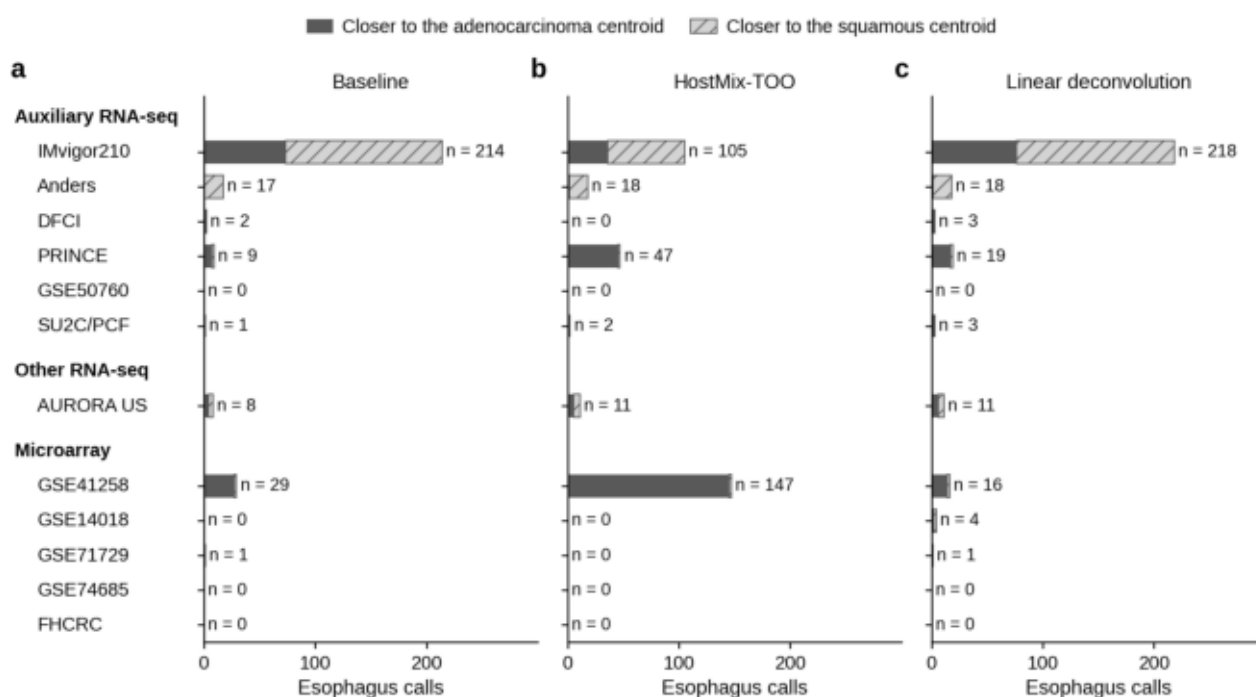


Figure S5. Assignment of esophagus calls to adenocarcinoma or squamous centroids

Each row is one cohort, grouped as the six auxiliary RNA-seq cohorts, the other RNA-seq cohort (AURORA US) and the five microarray cohorts, in the same order in every panel: a baseline, b HostMix-TOO, c linear deconvolution. Each esophagus call is assigned to the adenocarcinoma or the squamous centroid, whichever has the greater Spearman correlation with the profile. The n at the end of a bar is that panel's number of esophagus calls, including n = 0 when the classifier made none; GSE50760 had none under any classifier. Of 147 HostMix-TOO esophagus calls in GSE41258, 145 were closer to the adenocarcinoma centroid and 2 to the squamous centroid; in IMvigor210, 69 of 105 were closer to the squamous centroid. This assignment is post hoc.

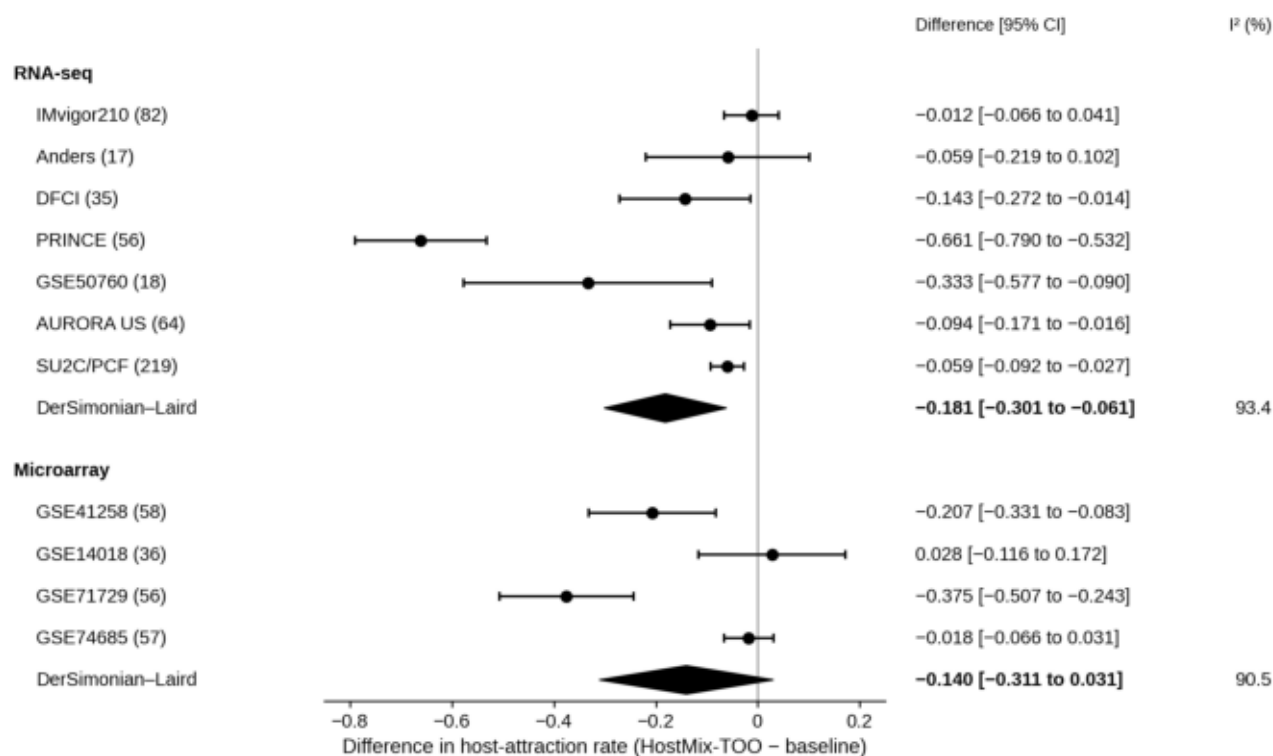


Figure S6. Cohort-level differences in host-attraction rate

Difference in host-attraction rate, HostMix-TOO minus baseline, in each cohort with at least 10 at-risk biopsies (at-risk n in parentheses), with Wald 95% intervals from the recorded variances. Diamonds show the DerSimonian-Laird random-effects estimate and its 95% interval for each layer (RNA-seq: seven cohorts, including the descriptive cohort AURORA US; microarray: four cohorts); the columns on the right give each difference with its 95% interval and the I^2 of each layer as a percentage. Cohort-level differences use each cohort's own one-sample-per-patient set, so patients profiled in both GSE74685 and FHCRC are counted in each cohort, and the at-risk n of GSE74685 (57) is larger than in Table 1 (49). FHCRC is absent because its at-risk count was below 10. The cohort differences and layer estimates were specified in advance as descriptive analyses; they do not replace the preregistered tests.